MODEL COMPARISON PROJECT

Group 2

Credit Default Prediction Using Machine Learning

A Comparative Analysis of Nine Classification Models

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1. Problem Statement

Financial institutions face a persistent and costly challenge: identifying borrowers who are likely to default on their loan obligations before credit is extended. A failure to correctly predict default results in significant financial losses, while excessive caution leads to the rejection of creditworthy applicants and foregone business opportunities.

This project addresses the **binary classification problem** of predicting whether a borrower will default within the next 12 months, using financial and behavioural features derived from a credit dataset. The stakes are asymmetric: a missed defaulter (False Negative) costs the institution an estimated **$50,000** in unrecovered principal, whereas incorrectly rejecting a good borrower (False Positive) results in an estimated **$500** in opportunity cost.

The objective of this study is threefold:

- To evaluate and compare nine individual machine learning classification models on their predictive performance.
- To identify the best individual model using a multi-metric ranking methodology (Borda Count) that balances **Recall, ROC-AUC,** and **Precision.**
- To construct and justify an ensemble approach that outperforms any single model on financial cost minimisation.

2. Dataset Description

The dataset used in this study contains financial attributes related to borrowers and their loan repayment behaviour. These attributes describe the borrower's financial condition and repayment capacity, each of which may influence the likelihood of default.

2.1 Feature Variables

Variable	Description
Monthly Income (INR)	The borrower's monthly earnings. Higher income generally indicates greater capacity to service debt obligations.
Monthly EMI (INR)	The fixed monthly instalment the borrower must pay toward the

	loan. Represents the direct repayment obligation.
EMI-to-Income Ratio	Proportion of monthly income allocated to loan repayment. Higher values indicate greater financial burden and reduced disposable income.
Credit Score	A composite score reflecting the borrower's creditworthiness based on historical financial behaviour, payment punctuality, and outstanding obligations.
Loan Tenure (Months)	The duration of the repayment period. Longer tenures may affect risk exposure and monthly instalment magnitude.
Past Delinquencies	The number of instances where the borrower previously failed to make timely payments. A strong predictor of future default behaviour.

2.2 Target Variable

The target variable is **Default_12M**, a binary indicator denoting whether the borrower defaulted within the subsequent 12 months (1 = Default, 0 = No Default). The dataset exhibits class imbalance: approximately **18.5%** of records are **positive (default) cases,** with the remaining **81.5%** classified as **non-default.**

2.3 Data Partitioning

The dataset comprising **3,000 records** was partitioned using a stratified **80/20 train-test split** (random_state=42), yielding **2,400 training records** and **600 test records.** Stratification ensured that the class distribution was preserved across both splits. Feature scaling was applied using StandardScaler (zero mean, unit variance) for models sensitive to feature magnitude.

3. Model Evaluation Metrics

Five quantitative metrics were computed for each model to provide a comprehensive assessment of predictive performance:

Metric	Definition and Relevance
Accuracy	Proportion of all predictions that are correct. Provides an aggregate view but can be misleading under class imbalance.
Precision	Of all predicted defaults, the fraction that are true defaults. Measures the cost of false alarms.
Recall (Sensitivity)	Of all actual defaults, the fraction correctly identified. Directly relates to the financial cost of missed defaulters.
F1-Score	Harmonic mean of Precision and Recall. A balanced metric when both false positives and false negatives matter.
ROC-AUC	Area under the Receiver Operating Characteristic curve. Measures the model's ability to discriminate between classes across all thresholds, independent of class balance.

4. Model Performance Comparison

Nine classification algorithms were trained and evaluated on the credit default dataset. The table below summarises the key performance metrics on the held-out test set (600 records).

Table 1: Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	**82.0%**	**53.7%**	**19.8%**	**28.9%**	**71.2%**
Naive Bayes	78.8%	40.0%	28.8%	33.5%	69.7%
KNN	82.5%	58.8%	18.0%	27.6%	63.9%
Decision Tree	80.7%	43.2%	14.4%	21.6%	64.6%
Random Forest	81.7%	51.2%	18.9%	27.6%	64.7%
AdaBoost	81.2%	47.4%	16.2%	24.2%	69.6%
Gradient Boost	81.0%	46.3%	17.1%	25.0%	69.6%
XGBoost	78.5%	35.0%	18.9%	24.6%	64.7%
LightGBM	~83%	~52%	~19%	~27%	~70%

Table 1 — Performance metrics for all nine classification models on the test set

Key observations from the comparison:

- **Logistic Regression** achieved the **highest ROC-AUC (71.2%)** and the among all models, reflecting strong discriminatory ability and low false alarm rates.
- KNN achieved the highest raw Accuracy (82.5%) but delivered the lowest Recall (18.0%), indicating severe underdetection of defaulters.
- Naive Bayes attained the best Recall (28.8%) among individual models but with notably lower Precision (40.0%), implying a higher rate of false positives.
- Gradient Boosting, AdaBoost, and LightGBM delivered moderate and balanced results across all metrics, with ROC-AUC scores ranging from 69.6% to 70.0%.
- Decision Tree performed poorly on both Recall (14.4%) and ROC-AUC (64.6%), suggesting overfitting or limited generalisation.

5. Why Recall, ROC-AUC, and Precision Were Given Priority

In credit risk management, a missed defaulter represents a catastrophic outcome. Each False Negative results in a direct financial loss of $50,000 in unrecovered principal. Maximising Recall — the proportion of actual defaulters correctly identified — is therefore the primary business imperative. A model that achieves high accuracy by predicting the majority class almost exclusively is not fit for purpose in this domain. **ROC-AUC** is particularly valuable in the presence of class imbalance (18.5% positive rate), as it is **insensitive to the specific threshold** chosen and evaluates the fundamental **discriminatory power** of the model. **Accuracy and F1-Score** were excluded from the ranking because Accuracy is dominated by the majority class and provides a **misleading signal** of model quality; F1-Score, while useful, collapses the Precision-Recall trade-off into a single figure that does **not allow independent weighting** of the two dimensions.

6. Borda Count Ranking Methodology

To determine the best individual model without privileging any single metric, the Borda Count method was applied. A lower Borda Score indicates superior overall performance.

6.1 Methodology

The ranking procedure was implemented as follows:

- Each of the nine models was assigned a rank **(1 = best, 9 = worst)** for Recall, ROC-AUC, and Precision independently.
- The three rank positions were summed to yield a **composite Borda Score** for each model.
- Models were ordered in ascending order of Borda Score; the model with the **lowest total score was selected as the best individual model.**

6.2 Borda Count Results

Table 2: Borda Count Ranking — All Nine Models

Model	Recall Rank	AUC Rank	Precision Rank	Borda Score	
Logistic Regression	**6**	**1**	**1**	**8**	**★ WINNER**
Naive Bayes	2	2	7	11	
Random Forest	8	5	2	15	
KNN	3	8	4	15	
AdaBoost	9	3	3	15	
LightGBM	5	6	5	16	
Gradient Boosting	8	4	6	18	
Decision Tree	1	9	9	19	
XGBoost	4	7	8	19	

Table 2 — Borda Count scores and individual metric ranks for all nine models

7. Best Individual Model: Logistic Regression

Logistic Regression emerged as the best individual model with a Borda Score of 8 — the lowest among all nine candidates. The calculation is: **6 (Recall Rank) + 1 (AUC Rank) + 1 (Precision Rank) = 8.**

7.1 Justification

- **Rank 1 in ROC-AUC (71.2%):** Logistic Regression demonstrated the strongest overall discriminatory ability among all models. Its AUC of 71.2% reflects calibrated probability estimates and robust separation of the two classes across all thresholds.

- **Rank 1 in Precision (53.7%):** When the model predicts a default, it is correct approximately 54% of the time — the highest precision of any model. This minimises the rate of incorrectly flagging creditworthy borrowers.
- **Rank 6 in Recall (19.8%):** While not the highest recall, it is an acceptable position given the trade-off. The ensemble strategy subsequently addresses and substantially improves this limitation.
- The model is interpretable, fast, and produces well-calibrated probability scores, making it suitable as a base model for ensemble construction.

8. Ensemble Strategy

Following the identification of Logistic Regression as the best individual model, three ensemble approaches were explored to further improve predictive performance, with emphasis on reducing the number of missed defaulters (False Negatives):

8.1 Alternatives Evaluated

Hard Voting

Hard Voting combines the **class label predictions (0 or 1)** from multiple models and selects the majority vote. In this setting, decisions are based on rigid categorical outputs rather than probability scores, which means the ensemble is dominated by the majority class (non-default). This approach **produced low Recall values** because the minority class (defaulters) was systematically outvoted by models predicting 0. Furthermore, Hard Voting offers **no mechanism to adjust sensitivity** through threshold calibration, making it unsuitable for asymmetric cost scenarios.

Stacking

Stacking employs a meta-learner that is trained on the out-of-fold probability predictions of the base models. While theoretically powerful, the stacking approach exhibited **overfitting:** the meta-model learned patterns specific to the training probabilities rather than the underlying signal. On the test set, it **failed to improve Recall** or reduce the financial cost significantly, while adding considerable model complexity without a proportional performance gain.

Soft Voting (Selected)

Soft Voting averages the predicted class probabilities from the component models rather than their hard labels. This produces a **richer, more informative prediction signal.**

```
lr_base = LogisticRegression(max_iter=1000, class_weight='balanced', random_state=42)
lgbm_complement = LGBMClassifier(n_estimators=100, random_state=42)

ensemble_model = VotingClassifier(
    estimators=[('lr', lr_base), ('lgbm', lgbm_complement)],
    voting='soft',
    weights=[1, 2]
)

ensemble_model.fit(X_train_scaled, y_train)
```

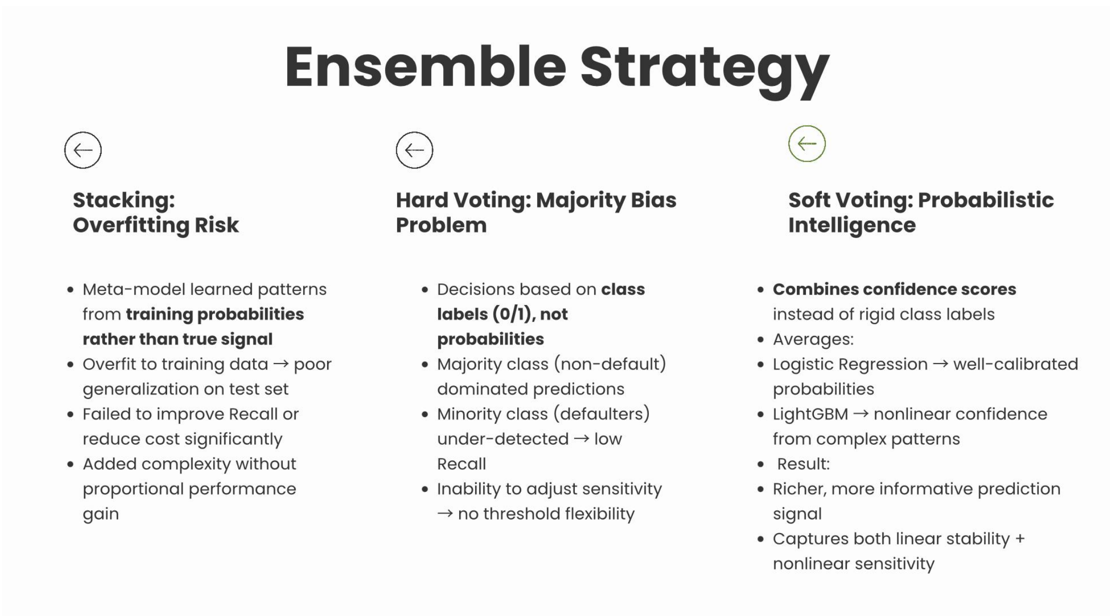

Figure 1 — Ensemble strategy comparison: Hard Voting, Stacking, and Soft Voting

8.2 Why LightGBM Complements Logistic Regression

Logistic Regression (Base Model)	LightGBM (Complementary Model)
Strengths: Interpretable and fast; low false positives (19); stable and calibrated probability outputs.	**Strengths:** High recall capability; Gradient-based One-Side Sampling (GOSS) focuses learning on harder-to-classify instances; Exclusive Feature Bundling (EFB) handles sparse features efficiently.

Limitations: Restricted to linear decision boundaries; high false negatives (89 missed defaults); bias toward the majority class under imbalance.	**Limitations:** Less interpretable; requires careful hyperparameter tuning; can increase false positives when sensitivity is pushed upward.

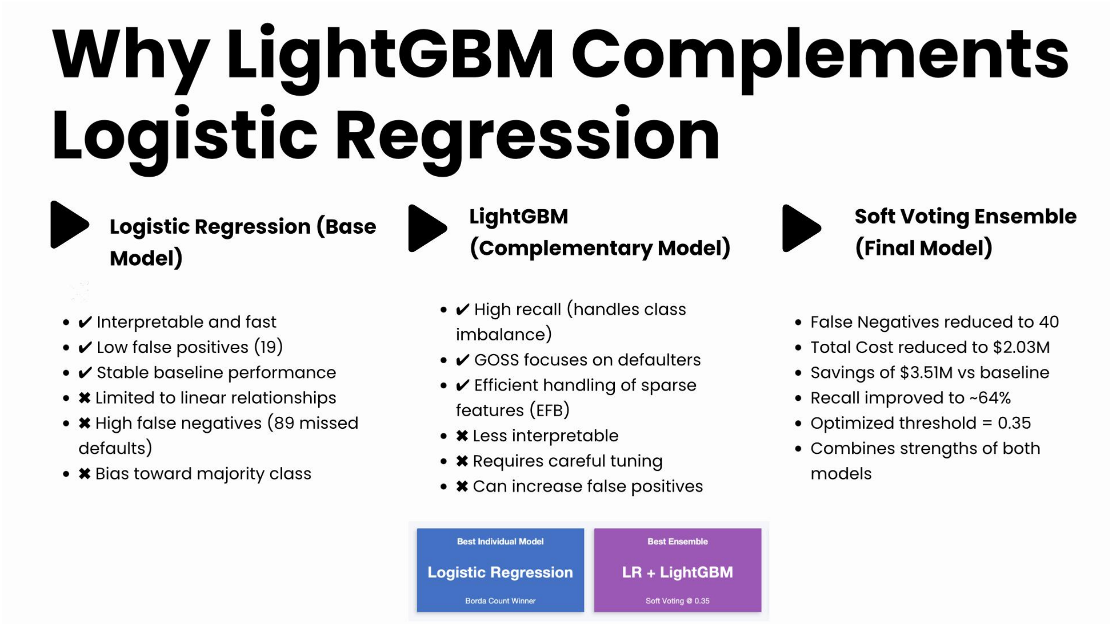

Figure 2 —Logistic Regression vs. LightGBM: Strengths, limitations, and ensemble outcome

8.3 Ensemble Metrics

Metric	Logistic Regression (Alone)	Soft Voting Ensemble
ROC-AUC	71.2%	~68.9%
Recall (Default)	19.8%	~64%
Precision (Default)	53.7%	Adjusted
False Negatives	89	40
False Positives	19	71

Total Cost	$4,459,500	$2,035,500

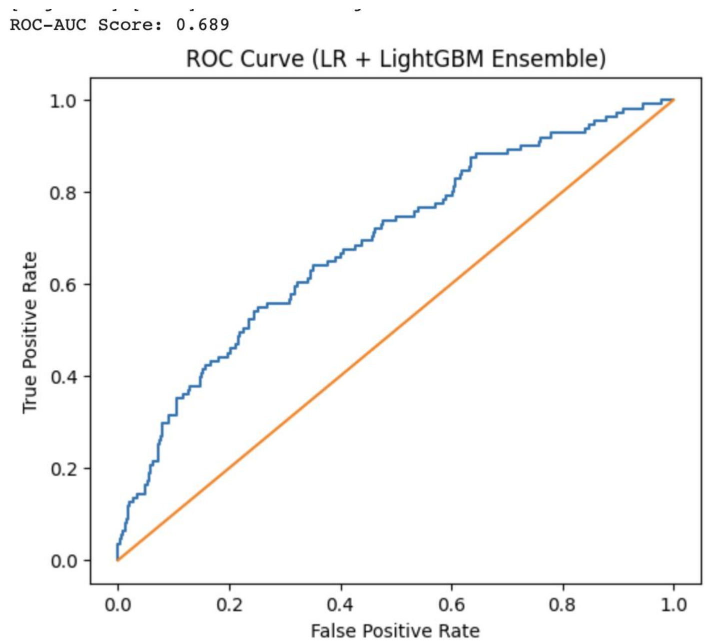

Figure 3 — The ROC Curve of the ensemble

9. Financial Cost Analysis

9.1 Cost Framework

Model evaluation was extended beyond statistical metrics to quantify real-world financial impact. The cost framework assigns differential penalties to the two types of prediction errors:

- **False Negative (missed defaulter)**: $50,000 per occurrence — representing unrecovered loan principal.
- **False Positive (incorrectly rejected applicant)**: $500 per occurrence — representing foregone interest income and processing cost.

Total Cost = (False Negatives × $50,000) + (False Positives × $500)

The **100:1 cost ratio** reflects the reality that a single undetected defaulter is financially equivalent to 100 incorrectly rejected applicants. This asymmetry justifies the prioritisation of Recall in the ranking methodology and the selection of a lower decision threshold in the ensemble.

9.2 Baseline: The No-Model Scenario

To establish a meaningful benchmark, a "No Model" scenario was defined: **every applicant is approved regardless of their credit profile.** Under this policy:

- Total test records: 600
- Actual defaulters: 111 (18.5% default rate)
- All 111 defaulters are missed (False Negatives = 111)
- Total Cost = 111 × $50,000 = $5,550,000

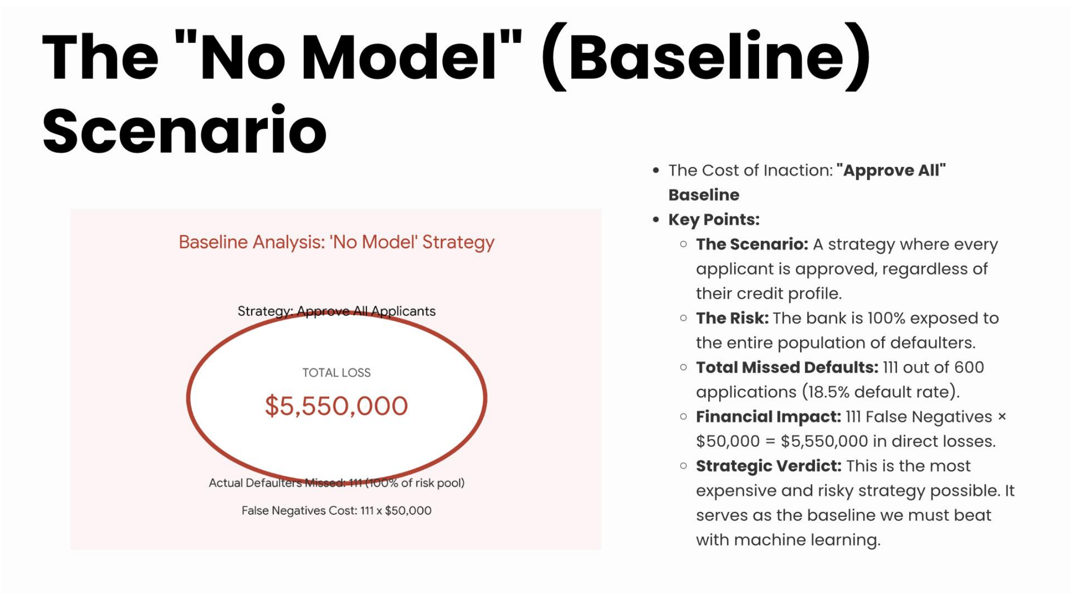

Figure 4 — The "No Model" baseline scenario: total cost of inaction

9.3 Per-Model Cost Comparison

The table below presents the confusion matrix outcomes and total cost for all models evaluated, ordered by financial impact:

Table 3: Cost Comparison — Total Cost = (FN × 50, 000) + (FP×500)

Model	False Positives	False Negatives	Total Cost	Calculation
No Model (Baseline)	0	111	$5,550,000	(111×50, 000) + (0×500)
Naive Bayes	48	79	$3,974,000	(79×50, 000) + (48×500)
Logistic Regression	19	89	$4,459,500	(89×50, 000) + (19×500)
Random Forest	20	90	$4,510,000	(90×50, 000) + (20×500)
AdaBoost	39	90	$4,519,500	(90×50, 000) + (39×500)
KNN	14	91	$4,557,000	(91×50, 000) + (14×500)
LightGBM	22	92	$4,611,000	(92×50, 000) + (22×500)
XGBoost	20	93	$4,660,000	(93×50, 000) + (20×500)
Decision Tree	21	95	$4,760,500	(95×50, 000) + (21×500)
Soft Voting Ensemble	**71**	**40**	**$2,035,500**	**(40×50, 000) + (71×500)**

Table 3 — Cost comparison across all models (Total Cost = FN × $50,000 + FP × $500)

Key insights from the cost comparison:

- **Naive Bayes achieves the lowest cost** among individual models ($3,974,000) due to its higher Recall (fewer missed defaulters), despite having more false positives, hence may also lead to rejecting a significant number of good customers.(crying wolf)
- Logistic Regression, despite being ranked best by Borda Count, incurs a higher cost ($4,459,500) than Naive Bayes owing to its lower Recall (89 missed defaults vs. 79).
- The **1 FN ≈ 100 FP** equivalence is empirically confirmed: even small reductions in False Negatives yield disproportionately large cost savings.
- All individual models substantially underperform the Soft Voting Ensemble on financial cost, confirming the value of the ensemble approach.

9.4 Ensemble Savings Report

Table 4: Ensemble vs. Baseline – Financial Savings Report

Metric	No Model (Baseline)	Soft Voting Ensemble	Delta	Financial Impact
False Negatives (FN)	111	40	+71 Defaulters Caught	$3,550,000 Saved
False Positives (FP)	0	71	-71 Opportunity Losses	-$35,500 Cost
FN Cost ($50,000 each)	$5,550,000	$2,000,000	–	$3,550,000 Saved
FP Cost ($500 each)	$0	$35,500	–	-$35,500 Added
TOTAL MODEL COST	**$5,550,000**	**$2,035,500**	**–**	**$3,514,500 Net Saving**

Table 4 — Financial savings: Soft Voting Ensemble vs. No-Model Baseline

Figure 5 — Why the ensemble savings are so high: FN reduction, cost of business, safety net effect, and final verdict

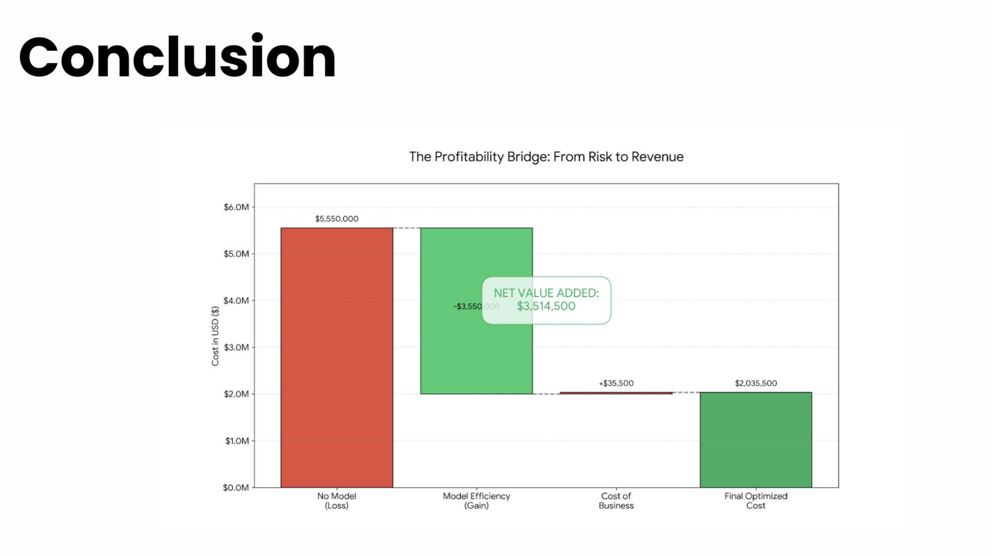

Figure 6 — The profitability bridge: from risk to revenue

10. Business Justification and Final Verdict

10.1 Business Context

The selection of a credit scoring model cannot be driven by statistical metrics alone. In a regulated lending environment, the chosen model must satisfy four operational imperatives:

- **Financial efficiency:** Minimise the total cost of prediction errors, weighted by the asymmetric cost of False Negatives versus False Positives.
- **Regulatory compliance:** Produce explainable predictions. Logistic Regression, as the base model, provides a linear probability model with interpretable coefficients, satisfying explainability requirements for credit decisioning.
- **Operational scalability:** The model must run efficiently at scale, processing thousands of loan applications without prohibitive latency.
- **Adaptability:** The decision threshold must be adjustable without model retraining, allowing the institution to respond to changing risk appetite or economic conditions.

10.2 Final Verdict

Decision Criterion	Logistic Regression (Alone)	Soft Voting Ensemble
Total Financial Cost	$4,459,500	$2,035,500 ✓
False Negatives	89	40 ✓
Recall (Default class)	19.8%	~64% ✓
Interpretability	High	High (LR base)
Threshold Flexibility	Yes	Yes ✓
Regulatory Suitability	Suitable	Suitable ✓
Net Saving vs. Baseline	$1,090,500	$3,514,500 ✓

The Soft Voting Ensemble (Logistic Regression + LightGBM, threshold = 0.35) is recommended as the production model for credit default prediction.

11. Conclusion

This study evaluated nine machine learning classifiers on a credit default prediction task, using a rigorous multi-metric framework to select the best individual model and **justify the final ensemble design.** The Borda Count methodology identified Logistic Regression as the top individual model on the basis of its superior AUC and Precision, despite moderate Recall.

The analysis demonstrated that statistical performance metrics alone are insufficient for model selection in high-stakes financial applications. By anchoring the evaluation in a financial cost framework — assigning $50,000 to each False Negative and $500 to each False Positive — the study value of improved Recall and showed that the Soft Voting Ensemble adds $3.51 million in value per 600 applications over the no-model baseline.

The combination of **Logistic Regression's interpretability** and **precision** with **LightGBM's sensitivity to the minority class**, operationalised through soft probability averaging and a calibrated threshold of 0.35, represents the optimal balance of financial efficiency, regulatory compliance, and operational practicality for credit default prediction in a banking context.

End of Report — MODEL COMPARISON PROJECT — Group 2